The Nyquist Criterion and Stability Margins

Contents

1. Overview . . . 2
2. The loop transfer function and the critical point . . . 4
   - 2.1 Cutting the loop open . . . 4
   - 2.2 Where the closed loop has its poles . . . 6
   - 2.3 A sine that comes back unchanged . . . 7
3. Counting poles from a walk: the argument principle . . . 9
   - 3.1 Mapping one point . . . 9
   - 3.2 Walking round a closed path . . . 11
4. The Nyquist criterion . . . 13
   - 4.1 The Nyquist contour: a walk round the right half-plane . . . 13
   - 4.2 The Nyquist plot of the heading loop . . . 14
   - 4.3 Counting turns round the critical point . . . 15
5. When a Bode plot misleads: a balancing robot . . . 16
6. Stability margins: how far from the critical point . . . 18
   - 6.1 Gain margin . . . 19
   - 6.2 Phase margin . . . 21
   - 6.3 Delay margin . . . 22
   - 6.4 The margins on the Bode plot . . . 23
7. Where margins and delay show up in real stacks . . . 24
8. Summary . . . 26
9. Sources . . . 27
10. Key terms . . . 28

Where this fits:

- **Builds on:** Block-diagram algebra.
- **Leads to:** Sensitivity functions; Ziegler–Nichols tuning.
- **Compare with:** Bode plot.

1. Overview

Key point: The robot's heading loop settles when the controller reads the heading without delay, but when its heading reaches the controller 0.25 s late, its swings grow instead of dying out. The Nyquist criterion tells a stable loop from an unstable one using only the loop's frequency response, delay included, and the stability margins say how much more gain, phase lag or delay the loop can take: 0.19 s of delay for our loop.

The robot can already turn on command. Its heading loop is a **PD control** (G-2962) loop, which turns the robot with a torque in proportion to the heading error and brakes it in proportion to the turn rate, driving a yaw inertia of 0.1 $\mathrm{kg \cdot m^2}$ with at most 0.2 $\mathrm{N \cdot m}$ of torque. RO-060 tuned the heading loop to $K_p = 0.2$ $\mathrm{N \cdot m/rad}$ and $K_d = 0.283$ $\mathrm{N \cdot m \cdot s/rad}$, plus a small integral gain of 0.05 against a dragging gearbox, so a turn never passes its target. Here we study the lightly damped first try of RO-061, $K_p = 0.4$ $\mathrm{N \cdot m/rad}$ and $K_d = 0.08$ $\mathrm{N \cdot m \cdot s/rad}$ with no integral action, because its small margins show how delay reduces them; Section 6.3 compares it with RO-060's tuned loop. So far:

- told to turn from 0 to 0.5 rad, it swings to 0.763 rad, 53 percent past the target, and settles, as its **step response** (G-2981) showed;
- its closed loop has the **poles** (G-3000), the values of the transfer function's complex variable s where it becomes infinite, at $-0.4 \pm 1.96\mathrm{j}$: complex numbers, with j the number whose square is -1, both 0.4 to the left of the imaginary axis, in the **left half-plane** (G-3003), so it is stable;
- its frequency response (G-3020), the gain (G-3021), how many times larger the output sine is, and the phase (G-3022), how far it is shifted, for a sine of each frequency, peaks at a gain of 2.55 near 1.92 rad/s.

All of this assumes the controller sees the heading the moment it changes. A real loop always has some delay. A sensor's filter, the computer and the motor driver each add time between a change of heading and the torque that answers it; PX4's tuning guide lists 1.9 to 8 ms for its gyro filter alone, and Autoware's speed controller compensates for 0.17 s of actuator delay (PX4 filter tuning; Autoware PID controller).

Figure 1 runs the same 0.5 rad turn three times, with the heading reaching the controller 0, 0.1 and 0.25 s late; nothing else changes. Watch the first swing: 0.763 rad with no delay, 0.865 rad (73 percent past the target) with 0.1 s. With 0.25 s the swings grow, 1.13, 1.48, 1.95 rad, and the motor's torque sits at its 0.2 $\mathrm{N \cdot m}$ limit from 1.25 s on.

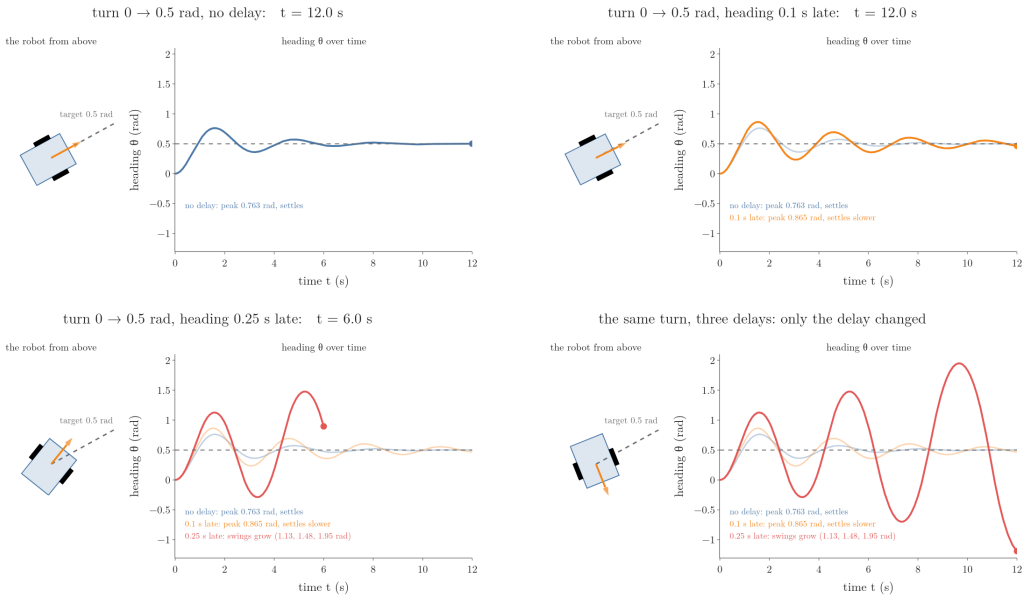


Figure 1: RO-061's turn from 0 to 0.5 rad (dashed line) with the same PD gains, run three times with the heading reaching the controller later. Left: the robot from above, turning. Right: the heading over time, earlier runs kept faint. No delay (blue): first swing to 0.763 rad, then it settles. 0.1 s late (orange): 0.865 rad, settles more slowly. 0.25 s late (red): the swings grow, 1.13, 1.48 and 1.95 rad. The last frame shows the three runs together

The opening question of this Note: **how much delay, or how much more gain, can the heading loop take before its swings stop dying out, and how can we tell from the loop alone, without solving for the closed loop's poles?**

Solving for the poles is no longer easy. A **time delay** (G-3012) multiplies a transfer function by an exponential of s, so the closed loop's equation is no longer a polynomial and the root formula for the delay-free loop does not apply. Even where it does, a pole's position says stable or unstable, but not how close to the edge the loop is or what to change (FBS §10.1). This Note answers the question in steps:

- the loop transfer function and the point -1, where the closed loop has its poles (Section 2);
- counting poles inside a closed path by walking round it, the argument principle (Section 3);
- the Nyquist criterion, which tells the 0.1 s loop (stable) from the 0.25 s loop (unstable) by counting turns round -1 (Section 4);
- why a Bode plot alone can mislead, on a balancing robot (Section 5);
- the gain, phase and delay margins: 0.19 s of delay for our loop (Section 6);
- where margins and delays show up in PX4 and Autoware (Section 7).

2. The loop transfer function and the critical point

> **Key point:** Cut the loop open and follow a signal once round: the product of everything it passes, here the PD controller times the heading plant times the delay, is the loop transfer function $L(s)$. The closed loop has a pole wherever $L(s) = -1$, and a pole sits on the edge of stability when a sine comes back once round the loop exactly as it went in.

2.1 Cutting the loop open

The delay of Figure 1 sits inside a loop: the heading goes from the robot to the controller, the controller's torque goes back to the robot, and round again. To study the whole loop at once, we cut it at one place and follow a signal once round (FBS §10.1).

Figure 2 cuts the loop at the torque. A signal put in at A turns the robot; the heading reaches the controller T_d seconds late; the comparison subtracts it; the controller turns it into a torque that comes out at B. Each block multiplies the signal's transfer function by its own. The product of everything once round, leaving out the minus sign of the comparison, is the **loop transfer function** (G-3035), $L(s)$ (FBS §10.1). The loop transfer function is also called the open-loop transfer function, because it is measured with the loop open.

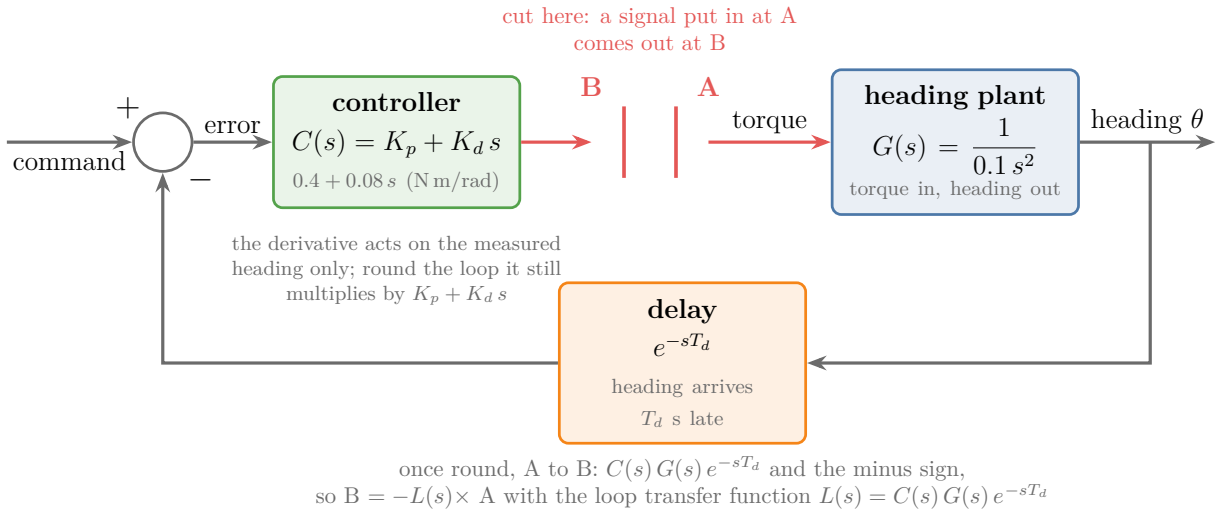


Figure 2: The heading loop cut open at the torque. From the command, the comparison forms the error, the PD controller C(s) = Kp + Kd s gives a torque, the heading plant G(s) = 1/(0.1 s^2) turns it into a heading, and the heading comes back through a delay of T_d seconds to the comparison with a minus sign. A signal put in at A goes once round and comes out at B multiplied by −L(s), with L(s) = C(s) G(s) e^(−s T_d)

The pieces, with:

- s, the complex variable of the Laplace transform, in 1/s, in which every transfer function of RO-062 is written;

- $C(s)$, the controller's transfer function, torque per radian of heading error, in $\mathrm{N \cdot m/rad}$;
- K_p, the proportional gain (G-2959), torque per radian of error, 0.4 $\mathrm{N \cdot m/rad}$;
- K_d, the derivative gain (G-2961), torque per radian per second of turn rate, 0.08 $\mathrm{N \cdot m \cdot s/rad}$;
- I, the robot's yaw **moment of inertia** (G-2955), its resistance to being turned, 0.1 $\mathrm{kg \cdot m^2}$;
- $G(s)$, the heading plant, heading per newton-metre of torque, in $\mathrm{rad/(N \cdot m)}$;
- T_d, the delay, in s;
- $L(s)$, the loop transfer function, radians of heading per radian, so it has no unit.

The plant's equation, the inertia times the turning acceleration equals the torque, becomes Is^2 times the heading equals the torque, because each derivative becomes a multiplication by s in a **transfer function** (G-2999). So:

$$C(s) = K_p + K_d\, s$$

$$G(s) = \frac{1}{Is^2}$$

$$L(s) = C(s)\, G(s)\, e^{-sT_d}$$

Without the delay, our numbers first, then the same steps in symbols:

$$L(s) = (0.4 + 0.08\, s) \times \frac{1}{0.1\, s^2}$$

$$= \frac{0.4 + 0.08\, s}{0.1\, s^2}$$

$$= \frac{0.8\, s + 4}{s^2}$$

$$L(s) = (K_p + K_d s) \times \frac{1}{Is^2}$$

$$= \frac{K_p + K_d\, s}{Is^2}$$

In RO-061's loop the derivative acts on the measured heading only, not on the command. Once round the loop, the controller still multiplies the heading by $K_p + K_d s$, so $L(s)$ is the same. With L known, the next step finds the closed loop's poles from it.

2.2 Where the closed loop has its poles

The loop transfer function $L(s)$ decides the closed loop's poles, which decide whether the swings of Figure 1 die out. With:

- θ, the robot's heading, in rad;
- r, the heading command, 0.5 rad in Figure 1;

and the loop closed with no delay, the heading is the plant times the torque, and the torque is K_p times the command minus $K_p + K_d s$ times the heading. The plant is:

$$G(s) = \frac{1}{0.1\,s^2} = \frac{10}{s^2}$$

Our numbers first:

$$\theta = \tfrac{10}{s^2}\Big(0.4\,r - (0.4 + 0.08s)\,\theta\Big)$$

$$\theta + \tfrac{0.8s+4}{s^2}\,\theta = \tfrac{4}{s^2}\,r$$

$$\theta\,(s^2 + 0.8s + 4) = 4\,r$$

$$\theta = \frac{4}{s^2 + 0.8\,s + 4}\,r$$

In symbols:

$$\theta = G\,(K_p r - C\,\theta)$$

$$\theta\,(1 + CG) = K_p G\,r$$

$$\theta = \frac{K_p G}{1 + L}\,r$$

The last line is the **closed-loop transfer function** (G-3007) from command to heading: RO-061's second-order loop, with **natural frequency** (G-2991) 2 rad/s and **damping ratio** (G-2992) 0.2. Its denominator is $1 + L(s)$, and every other signal of the loop has the same denominator (FBS §12.1). So the closed loop's poles are the values of s where:

$$1 + L(s) = 0$$

$$L(s) = -1$$

For our loop, multiplying by s^2:

$$s^2 + 0.8\,s + 4 = 0$$

$$s = -0.4 \pm \sqrt{0.16 - 4}$$

$$= -0.4 \pm 1.96\,j$$

At either pole, $L(s)$ is exactly -1 (notebook). The point -1 of the complex plane is the **critical point** (G-3036): the closed loop has a pole at every s that L sends there (FBS §10.1). With a delay, those s are hard to find; first we look at the edge of stability itself.

2.3 A sine that comes back unchanged

A pole on the imaginary axis gives a swing that neither grows nor dies: the edge between the 0.1 s run of Figure 1, which settles, and the 0.25 s run, which grows. On the imaginary axis $s = j\omega$, and $L(j\omega)$ is the frequency response of the loop cut open: a complex number whose length is the gain and whose angle is the phase the open loop gives a sine of frequency ω, in rad/s. So the loop is on the edge when, at some frequency,

$$L(j\omega) = -1$$

a gain of 1 and a phase of $-180°$ at the same frequency.

A microphone held near its own loudspeaker howls for the same reason that -1 marks the edge: the sound goes round, comes back in step and at the same loudness, and keeps itself going. Put a sine in at A of Figure 2. Once round, it comes back at B as $-L(j\omega)$ times the sine, the minus sign coming from the comparison. If $L(j\omega) = -1$, the sine at B is the sine at A. We could then join A to B and take the input away: the loop would keep the sine going by itself (FBS §10.1).

Our loop has a gain of exactly 1 at 2.08 rad/s (Section 6.2 derives it). The frequency where the gain of L is 1 is the **gain crossover frequency** (G-3042), written ω_{gc} (FBS §10.3). There, with no delay, L has length 1 and angle $-157.4°$, so the sine comes back 22.6° early.

A delay takes away phase and leaves the gain alone. With:

- ω, the sine's frequency, in rad/s;
- T_d, the delay, in s;

the delay's factor at the frequency ω is a complex number of length 1:

$$e^{-j\omega T_d} \ : \ \text{length 1}$$

$$\text{angle } \ -\omega T_d \text{ (in radians)}$$

Figure 3 adds delay in steps of 0.02 s and shows the sine at A (dashed) and at B (solid). Watch B slide later with every step. At 0.19 s B lies on A, and on the right the point $L(j\omega)$ times the delay's factor has turned clockwise onto -1. The 0.19 s is the line between the settling and the growing runs of Figure 1.

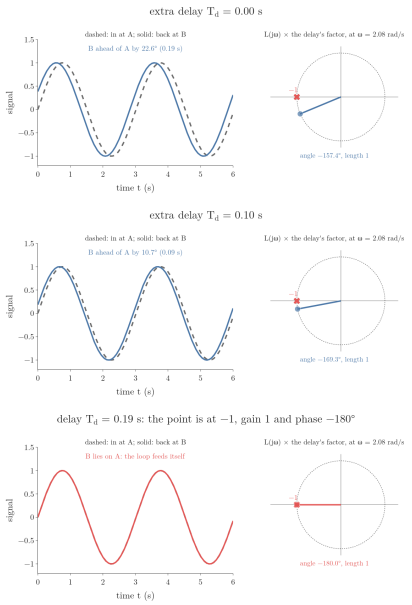


Figure 3: Once round the loop at $\omega = 2.08$ rad/s, where the gain of L is 1. Left: the sine put in at A (grey, dashed) and the sine that comes back at B (solid). Right: the loop's frequency response L(jω) times the delay's factor e^(−jω T_d), T_d the delay, on the circle of radius 1, with -1 marked. Each frame adds 0.02 s of delay. With no delay B is 22.6° (0.19 s) ahead of A; with 0.1 s, 10.7° ahead; at 0.19 s B lies on A and the point is at -1, angle $-180°$, length 1

One frequency at a time is not enough to decide stability: a pole can sit off the imaginary axis, and the delay's effect differs at every frequency. We need a way to count the closed loop's poles in the whole right half-plane, which needs one idea from complex numbers.

3. Counting poles from a walk: the argument principle

> **Key point:** A transfer function sends each complex number s to another complex number. When s walks clockwise round a closed path, the image turns round the origin once clockwise for every zero inside the path and once counter-clockwise for every pole inside: the argument principle.

3.1 Mapping one point

To find whether the closed loop has poles in the right half-plane without solving $1 + L(s) = 0$, we feed complex numbers into L and look at what comes out.

A complex number $a + bj$ is the point a across and b up, with $j^2 = -1$; its length is its distance from 0 and its angle is measured from the positive real axis. Multiplying two complex numbers multiplies their lengths and adds their angles:

$$(1 + j)(1 + j) = 1 + 2j + j^2 = 2j$$

$$\text{lengths: } \sqrt{2} \times \sqrt{2} = 2$$

$$\text{angles: } 45^\circ + 45^\circ = 90^\circ$$

Dividing divides the lengths and subtracts the angles.

Putting one point into L and getting one point out is called mapping the point; the output lives in its own plane, the w-plane. Take $s = -1+j$ in the heading loop's $L(s) = (0.8s+4)/s^2$, with no delay:

$$s^2 = (-1 + j)^2 = 1 - 2j + j^2$$

$$= -2j$$

$$0.8\,s + 4 = -0.8 + 0.8j + 4$$

$$= 3.2 + 0.8j$$

$$L(s) = \frac{3.2 + 0.8j}{-2j} = \frac{(3.2 + 0.8j)\,j}{-2j \times j}$$

$$= \frac{3.2j - 0.8}{2}$$

$$= -0.4 + 1.6j$$

The same answer comes from arrows (Figure 4). Written with its zero and its poles:

$$L(s) = \frac{0.8\,s + 4}{s^2} = 0.8\,\frac{s + 5}{s^2}$$

L has a zero at -5 and a double pole at 0. Draw an arrow from the zero to s and from the pole to s. The length of $L(s)$ is 0.8 times the zero's arrow divided by the pole's arrow twice; its angle is the zero's angle minus the pole's angle twice (FBS §10.2):

$$\text{length} = 0.8 \times 4.123/1.414^2$$

$$= 1.649$$

$$\text{angle} = 14.0^\circ - 2 \times 135.0^\circ$$

$$= -256.0^\circ$$

An angle of -256.0° points the same way as 104.0°, the angle of $-0.4 + 1.6$j, whose length is 1.649. One point maps to one point; a closed path of points maps to a closed curve.

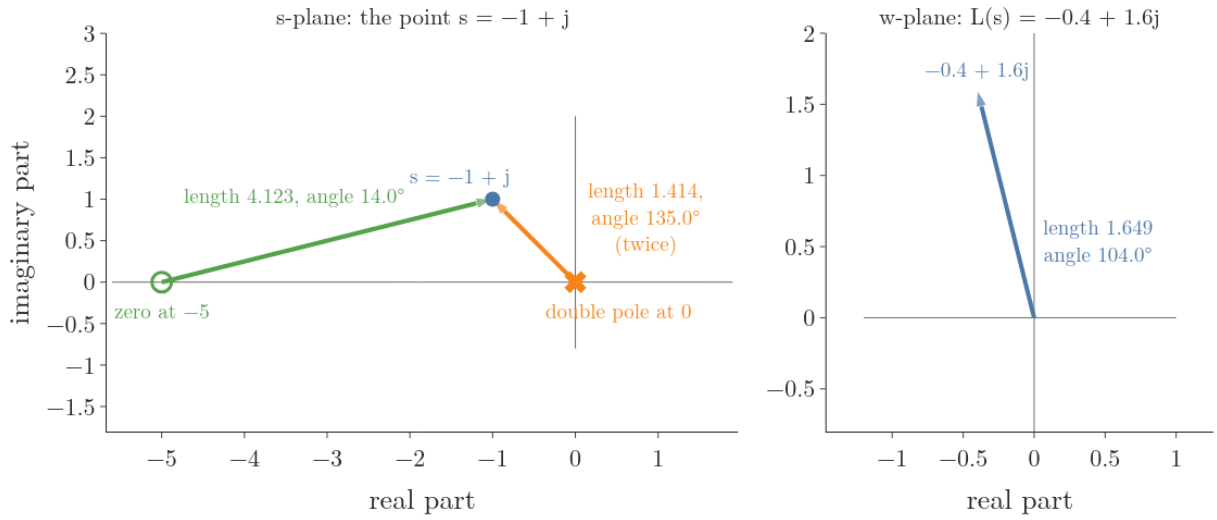


Figure 4: One point mapped through L(s) = 0.8 (s + 5)/s^2. Left, the s-plane: the zero of L at −5 (green circle), its double pole at 0 (orange cross) and the point s = −1 + j, with the arrow from the zero (length 4.123, angle 14.0°) and from the pole (length 1.414, angle 135.0°, counted twice). Right, the w-plane: L(−1 + j) = −0.4 + 1.6j, length 1.649, angle 104.0°

3.2 Walking round a closed path

One point gives one number, but the opening question needs to know whether any closed-loop pole lies anywhere in the right half-plane. A walk round a closed path tells what lies inside the path. We walk round points of $1 + L(s)$, whose zeros are the closed-loop poles -0.4 ± 1.96j (Section 2.2) and whose poles are the double pole of L at 0, since:

$$1 + L(s) = \frac{s^2 + 0.8\,s + 4}{s^2}$$

Figure 5 walks s clockwise round four circles and draws the image $1 + L(s)$, with a running count of its clockwise turns round 0. Watch the count at the end of each circle:

- round the zero $-0.4 + 1.96$j, the image turns once clockwise: $+1$;
- round the double pole at 0, it turns twice counter-clockwise: -2;
- round nothing, it swings and comes back: 0;
- round both zeros and the double pole, two clockwise and two counter-clockwise turns cancel: 0.

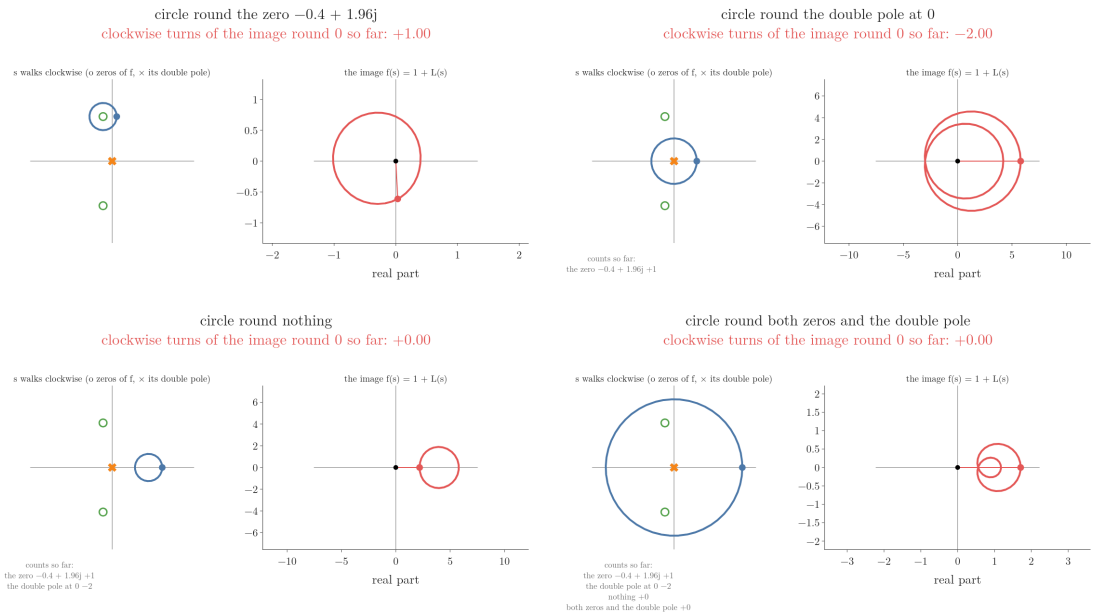


Figure 5: The argument principle on $1 + \mathrm{L(s)} = (\mathrm{s}^2 + 0.8\,\mathrm{s} + 4)/\mathrm{s}^2$. Left: s walks clockwise round a circle; green circles are the zeros $-0.4 \pm 1.96\mathrm{j}$ (the closed-loop poles), the orange cross the double pole at 0. Right: the image $1 + \mathrm{L(s)}$, with the count of its clockwise turns round 0 so far. Four circles: round the zero $-0.4 + 1.96\mathrm{j}$ $(+1)$, round the double pole (-2), round nothing (0), round both zeros and the double pole (0). The axes of the right panel change from circle to circle; the last frame lists the four counts

The reason lies in the arrows of Section 3.1. The angle of $1 + L(s)$ is the sum of the angles of the arrows from its zeros minus those from its poles. While s walks once clockwise round a zero, the arrow from that zero to s turns once clockwise, $-360°$. An arrow from a zero or a pole outside the path swings back and forth and ends where it started. An arrow from a pole inside also turns $-360°$, but a pole's angle is subtracted, so it turns the image $+360°$, counter-clockwise (FBS §10.2).

One full turn of a closed curve round a point is an **encirclement** (G-3038); the net number of turns, clockwise minus counter-clockwise, is the winding number. The rule that counts them is the **argument principle** (G-3037) (FBS Theorem 10.2). With:

- N, the number of clockwise turns of the image round 0, a whole number;
- Z, the number of zeros of the function inside the path;
- P_f, the number of poles of the function inside the path;

$$N = Z - P_f$$

The four circles of Figure 5, as turns = zeros − poles:

$$\text{round the zero: } 1 = 1 - 0$$

$$\text{round the pole:} \quad -2 = 0 - 2$$

$$\text{round nothing:} \quad 0 = 0 - 0$$

$$\text{round all four:} \quad 0 = 2 - 2$$

The count gives only the difference: to learn Z, the closed-loop poles, we must know how many poles lie inside.

4. The Nyquist criterion

> **Key point:** Draw $L(s)$ as s walks up the imaginary axis and round the right half-plane: if this Nyquist plot turns clockwise round -1 N times and L has P poles in the right half-plane, the closed loop has $N + P$ poles there. Our loop gives $N = 0$ with 0.1 s of delay (stable) and $N = 2$ with 0.25 s (two growing poles).

4.1 The Nyquist contour: a walk round the right half-plane

The closed loop is unstable if $1 + L(s)$ has a zero anywhere in the right half-plane, because a pole there makes the response grow. So the walk of Section 3.2 must go round the whole right half-plane. The walk goes up the imaginary axis from far below to far above and back down round a half circle so large that it holds every pole and zero on the right (FBS §10.2).

The walk must not pass through a pole of L, where L is infinite. Our $L(s) = (0.8s + 4)/s^2$ has its double pole at 0, on the imaginary axis, so the walk steps round it on a small half circle to its right. The double pole then lies outside, on the left of the walk. The closed walk, taken clockwise, is the **Nyquist contour** (G-3039) (FBS §10.2), shown in Figure 6.

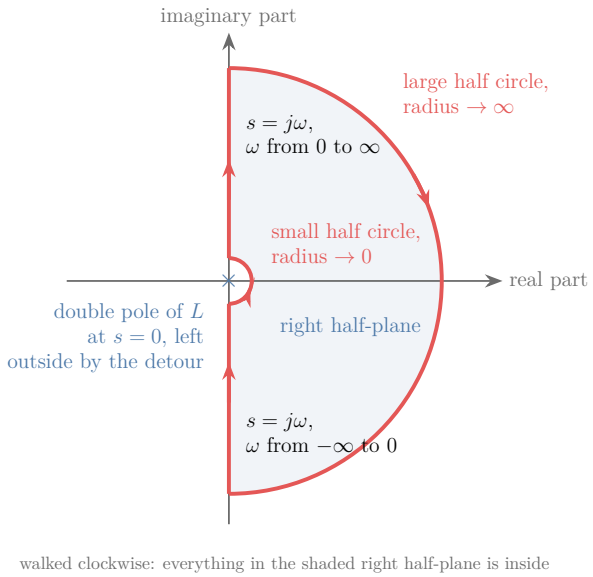


Figure 6: The Nyquist contour of the heading loop, walked clockwise: up the imaginary axis (s = jω, ω a frequency) from minus infinity to 0, round a small half circle that steps round the double pole of L at s = 0 on its right, up from $\omega = 0$ to infinity, and back round a large half circle. Everything in the shaded right half-plane is inside; the double pole at 0 is left outside

4.2 The Nyquist plot of the heading loop

To tell Figure 1's 0.1 s loop from its 0.25 s loop, we walk s along the Nyquist contour and draw $L(s)$; the curve is the **Nyquist plot** (G-3040) (FBS §10.2). Its four parts come from the four parts of the walk:

- on the imaginary axis above 0, $s = j\omega$, so this part is the frequency response of L, the loop cut open: one point per frequency, its length the gain and its angle the phase of the Bode plot (G-3023);
- on the imaginary axis below 0, the frequencies are negative and the curve is the mirror image of the first part across the real axis (FBS §10.2);
- on the large half circle, L shrinks to 0, since it falls like 0.8 divided by the length of s;
- on the small half circle, L is close to 4 divided by s^2, so its length is very large, and its angle, minus twice the angle of s, sweeps from +180° to −180°: one full clockwise turn, very far from the origin.

The last part goes off to infinity. To draw it, Figure 7 squeezes the plane: points within a distance 2 of the origin keep their places, farther points are pulled in, and infinity lands on a grey circle of radius 4. The squeeze slides points only along lines through the origin and leaves −1 and everything within 2 of the origin in place, so the curve never crosses −1 because of it and its turns round −1 do not change.

We count turns round −1, not round 0. The function whose zeros are the closed-loop poles is $1 + L$, and $1 + L$ is the curve of L shifted 1 to the right. Its turns round 0 are the turns of L

round -1 (FBS §10.2). Counting those turns is the last step.

4.3 Counting turns round the critical point

The count now decides Figure 1's two delayed runs. Apply the argument principle to $1 + L(s)$ on the Nyquist contour, with:

- N, the number of clockwise turns of the Nyquist plot of L round the critical point -1;
- P, the number of poles of L inside the contour, that is in the right half-plane;
- Z, the number of zeros of $1 + L$ in the right half-plane, which are the closed-loop poles there.

$1 + L$ has the same poles as L, because adding 1 does not change the denominator. So the argument principle gives, first for the 0.25 s loop, whose plot turns twice clockwise round -1 ($N = 2$, Figure 7 below) and whose L has no poles in the right half-plane ($P = 0$), then in symbols:

$$2 = Z - 0$$

$$Z = 2 + 0 = 2$$

$$N = Z - P$$

$$Z = N + P$$

The rule $Z = N + P$ is the **Nyquist stability criterion** (G-3041): the closed loop is stable when $Z = 0$, that is exactly when the Nyquist plot turns round -1 counter-clockwise once for every pole of L in the right half-plane (FBS Theorem 10.3). Our L has no poles in the right half-plane, its double pole at 0 being outside the contour, so $P = 0$ and the loop is stable exactly when $N = 0$.

Figure 7 traces the Nyquist plot of the heading loop for the two delays of Figure 1. Picture a pin at -1 and a string from the pin to the moving point: the count is how many times the string winds clockwise round the pin (FBS §10.2). Watch the frequency-response half (ω from 0 up) pass -1:

- 0.1 s: it comes in from far left just below the real axis and passes below -1, half a turn counter-clockwise (-0.5); its mirror adds -0.5; the far circle adds $+1$. So $N = 0$ and $Z = 0$: stable.
- 0.25 s: the delay has turned it, so it comes in just above the real axis and passes above -1, half a turn clockwise ($+0.5$); the mirror adds $+0.5$; the far circle $+1$. So $N = 2$ and $Z = 2$: two closed-loop poles in the right half-plane.

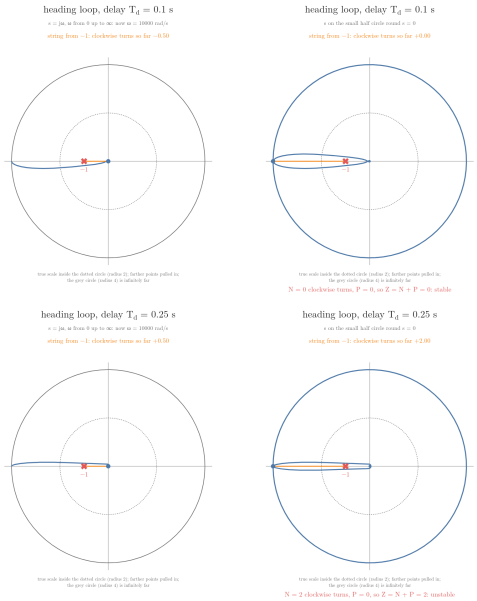


Figure 7: The Nyquist plot of the heading loop L(s) = (0.8 s + 4) eˆ(−s T_d)/s^2, traced as s walks the Nyquist contour, with a string (orange) from −1 to the moving point and the count of its clockwise turns. First T_d = 0.1 s, then T_d = 0.25 s. True scale inside the dotted circle of radius 2; farther points are pulled in so that infinity lands on the grey circle of radius 4. The last frame of each delay, with N the clockwise turns round −1, P the poles of L in the right half-plane and Z the closed-loop poles there: N = 0, P = 0, so Z = 0 (stable); N = 2, P = 0, so Z = 2 (unstable)

The two poles are a pair, 0.139 ± 2.064j for the 0.25 s loop (notebook): their positive real part makes the swing grow, and their imaginary part, 2.064 rad/s, gives a period of 3.04 s. The pair makes the growing swings of Figure 1, where the torque limit later stretches them. For the 0.1 s loop the pair is −0.205 ± 2.054j: still on the left, decaying more slowly than −0.4 ± 1.96j without delay.

The criterion needs P (FBS §10.2). For our loop $P = 0$ is clear, because the heading plant has no pole in the right half-plane. A robot whose plant has such a pole needs encirclements to be stable, and for it the Bode-plot reading fails.

5. When a Bode plot misleads: a balancing robot

> **Key point:** The Bode-plot reading "unstable if the gain is above 1 where the phase is −180°" holds only when L has no poles in the right half-plane, like our heading loop. A balancing robot's loop has one, and only the Nyquist count judges its stability correctly.

So far we could judge our loop from its Bode plot alone: where the gain is 1, is the phase above −180°? The reading treats each frequency as going round the loop on its own and asks whether it grows (FBS §10.1). The reading assumes the loop has nothing that grows by itself.

A two-wheeled robot that balances upright breaks that assumption: without control its tilt

grows on its own, so its plant has a pole in the right half-plane, which our heading loop does not. We use the textbook's version, with time measured in units that make the falling rate 1 (FBS Example 10.5). With:

- $G_b(s)$, the tilt plant, with poles at $+1$ (falling) and -1;
- $C_b(s)$, a PD controller;
- $L_b(s)$, its loop transfer function;

$$G_b(s) = \frac{1}{s^2 - 1}$$

$$C_b(s) = s + 2$$

$$L_b(s) = \frac{s + 2}{s^2 - 1}$$

At the frequency 0, the Bode plot reads:

$$L_b(0) = \frac{2}{-1} = -2$$

A gain of 2 at a phase of $-180°$: the reading says unstable. The closed loop's poles say otherwise:

$$1 + L_b(s) = 0$$

$$s^2 - 1 + s + 2 = 0$$

$$s^2 + s + 1 = 0$$

$$s = -0.5 \pm 0.866\, j$$

Both are on the left: the robot balances. The Nyquist count agrees. In Figure 8, watch the plot start at -2 and go once round -1 counter-clockwise, so $N = -1$. L_b has one pole in the right half-plane, so $P = 1$:

$$Z = N + P = -1 + 1 = 0$$

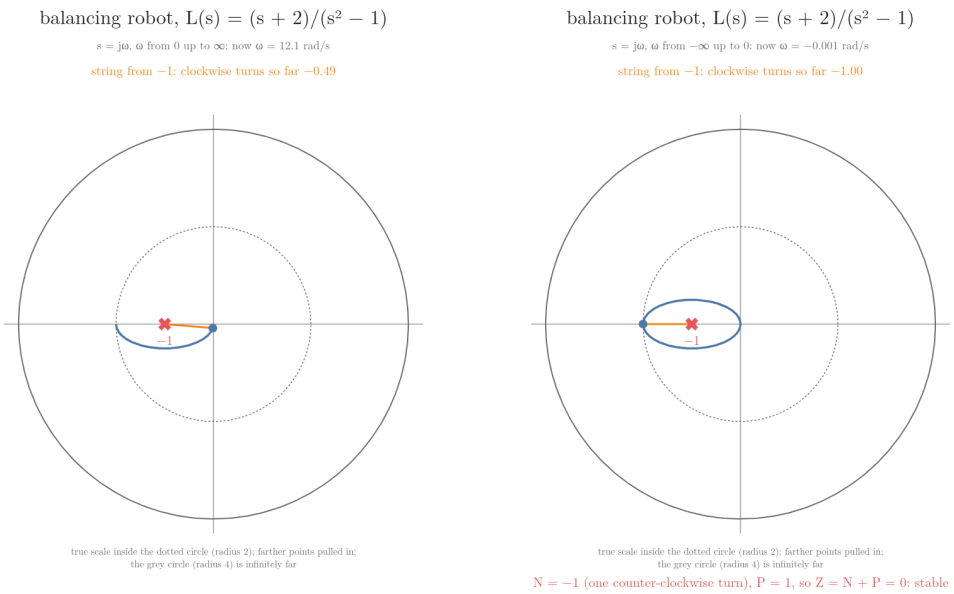


Figure 8: The Nyquist plot of the balancing robot's loop $L(s) = (s + 2)/(s^2 - 1)$, traced as s walks the Nyquist contour, with a string from -1 counting clockwise turns. The plot starts at $L(0) = -2$ and goes once round -1 counter-clockwise: N, the clockwise turns, is -1. With $P = 1$ pole of L in the right half-plane, the closed loop has $Z = N + P = 0$ poles there: stable

The Bode reading is a short cut that works for loops like our heading loop, with no poles of L in the right half-plane and one crossing of the gain 1. When those conditions fail, the Nyquist criterion is the test (FBS §10.3). With stability settled, the opening question remains: how much delay or gain can our loop take?

6. Stability margins: how far from the critical point

> **Key point:** The gain, phase and delay margins measure how far the Nyquist plot stays from -1 in extra gain, extra phase lag and extra delay, quantities we can change on the robot. Our delay-free loop has no gain limit, a phase margin of 22.6° and a delay margin of 0.19 s.

Figure 1's turns settle with no delay and with 0.1 s, but the Nyquist count showed the 0.25 s loop past the edge. A stable loop is not enough: we want to know how far from instability it is (FBS §10.3), measured in quantities that change on a real robot, such as a heavier robot, which lowers the gain, or a slower computer, which adds delay. Figure 9 reads all three margins off the Nyquist plot of the loop with 0.1 s of delay, the middle run of Figure 1.

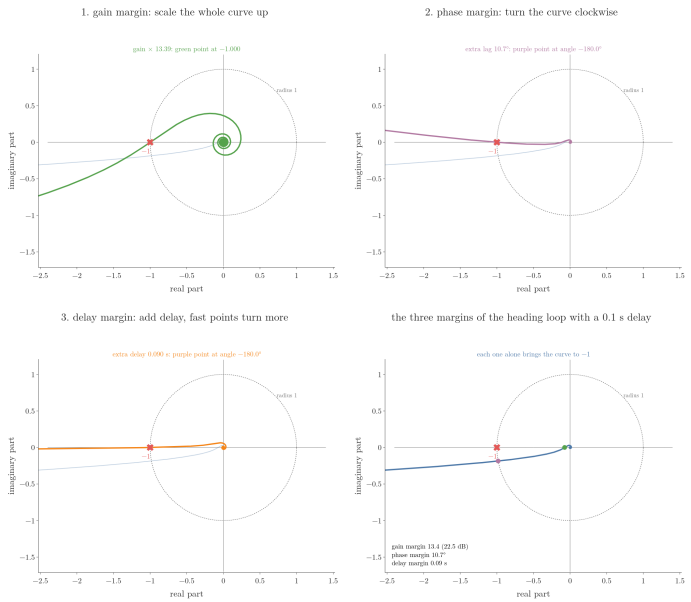


Figure 9: The three margins of the heading loop with 0.1 s of delay, read off the part of its Nyquist plot for ω above 0 (true scale; the dotted circle has radius 1; -1 marked in red). 1, gain margin: the curve is scaled up until the green point, where it crosses the negative real axis ($\omega = 11.66$ rad/s, at -0.0747), reaches -1: $\times$ 13.4. 2, phase margin: the curve is turned clockwise until the purple point, where it crosses the circle ($\omega = 2.08$ rad/s, angle $-169.3°$), reaches -1: 10.7°. 3, delay margin: extra delay turns each point by ω times the delay, faster points more; the purple point reaches -1 at 0.09 s. The last frame lists the three margins

6.1 Gain margin

The opening asked how much more gain the loop can take. Raising both controller gains by a factor k multiplies L by k and stretches the whole Nyquist plot away from 0 (FBS §10.3). The plot reaches -1 when the point where it crosses the negative real axis lands on -1. The crossing is where the phase of L, the loop transfer function of Section 2.1, is $-180°$; its frequency is the **phase crossover frequency** (G-3043), ω_{pc}. The factor that brings the crossing to -1 is the **gain margin** (G-3044) (FBS eq. 10.5). With:

- ω_{pc}, the phase crossover frequency, in rad/s;
- g_m, the gain margin, a pure number, or 20 times its base-10 logarithm in decibels;

$$g_m = \frac{1}{|L(j\omega_{pc})|}$$

Without delay, the phase of our L is the angle of $4 + 0.8j\omega$ minus twice the angle of $j\omega$:

$$\angle L(j\omega) = \arctan(0.2\,\omega) - 180°$$

The arctangent is above 0 for every positive frequency, so the phase never reaches $-180°$, the

plot never crosses the negative real axis, and no gain increase makes the loop unstable. A check on the poles: with the gains times k, the closed loop's equation is

$$s^2 + 0.8k\,s + 4k = 0$$

and its two poles add up to, and multiply to:

$$\text{sum of the poles} = -0.8\,k$$

$$\text{product of the poles} = 4\,k$$

A positive product means two real poles of the same sign, or a complex pair; a negative sum then puts both in the left half-plane, for every k above 0.

With 0.1 s of delay, the delay's angle grows with the frequency without end, so the phase, in radians, does reach $-\pi$:

$$\angle L = \arctan(0.2\,\omega) - \pi - 0.1\,\omega$$

The notebook finds the crossing at 11.66 rad/s. The gain there, with the frequency squared 135.96:

$$|L(j11.66)| = \frac{\sqrt{16 + 0.64 \times 135.96}}{135.96}$$

$$= \frac{\sqrt{103.01}}{135.96} = 0.0747$$

$$g_m = 1/0.0747 = 13.4$$

$$= 22.5 \text{ dB}$$

With 0.1 s of delay, the gains could grow 13.4 times before the loop oscillates; without delay, without limit. Gain is one direction away from -1; phase is the other.

6.2 Phase margin

The phase margin measures the other direction: how much extra phase lag the loop can take at the gain crossover frequency, where the loop transfer function L has length 1, before the point there turns onto -1. With:

- ω_{gc}, the gain crossover frequency, where the gain of L is 1, in rad/s;
- φ_m, the **phase margin** (G-3045), in degrees;

$$\varphi_m = 180^\circ + \angle L(j\omega_{gc})$$

(FBS eq. 10.6). For our loop without delay, the gain of $L(j\omega)$ is the length of $4 + 0.8j\omega$ divided by ω^2; setting it to 1, then squaring:

$$\frac{\sqrt{16 + 0.64\,\omega^2}}{\omega^2} = 1$$

$$16 + 0.64\,\omega^2 = \omega^4$$

$$\omega^2 = \frac{0.64 + \sqrt{0.64^2 + 64}}{2}$$

$$= \frac{0.64 + 8.026}{2} = 4.333$$

$$\omega_{gc} = 2.08 \text{ rad/s}$$

In symbols:

$$\frac{\sqrt{K_p^2 + K_d^2\,\omega^2}}{I\,\omega^2} = 1$$

$$K_p^2 + K_d^2\,\omega^2 = I^2\,\omega^4$$

The phase there:

$$\angle L = \arctan(0.2 \times 2.08) - 180^\circ$$

$$= 22.6^\circ - 180^\circ = -157.4^\circ$$

$$\varphi_m = 180^\circ - 157.4^\circ = 22.6^\circ$$

In symbols:

$$\varphi_m = \arctan(K_d\,\omega_{gc}/K_p)$$

With 0.1 s of delay the gain is unchanged, so the crossover is still 2.08 rad/s, but the delay subtracts an angle:

$$2.08 \times 0.1 = 0.208 \text{ rad} = 11.9^\circ$$

$$\varphi_m = 22.6^\circ - 11.9^\circ = 10.7^\circ$$

Typical designs keep the phase margin between 30° and 60° (FBS §10.3). Our 22.6° is below that range, in the same loop whose turn overshoots by 53 percent in RO-061. RO-060's tuned gains, $K_p = 0.2$ and $K_d = 0.283$, cross over at 2.91 rad/s with a phase margin of 76.4° (notebook). A delay is a phase lag that grows with the frequency, which turns the phase margin into a time.

6.3 Delay margin

A delay takes phase without changing the gain, so the delay that turns the point at ω_{gc} onto -1 is the one whose phase loss there equals the phase margin. The delay that does so is the **delay margin** (G-3046): the smallest extra delay that makes the loop unstable (FBS §10.3). With:

- T_m, the delay margin, in s;
- φ_m, the phase margin in radians, 22.6° = 0.3945 rad;
- ω_{gc}, the gain crossover frequency, 2.0815 rad/s;

the delay's phase loss at the crossover must equal the margin, our numbers first:

$$2.0815 \times T_m = 0.3945$$

$$T_m = 0.3945/2.0815 = 0.19 \text{ s}$$

$$\omega_{gc}\,T_m = \varphi_m$$

$$T_m = \varphi_m/\omega_{gc}$$

FBS §10.1 derives the same delay for a motor loop. MATLAB's `allmargin` follows the same rule: for its example it reports a phase margin of 29.11°, which is 0.5081 rad, at 1.7844 rad/s, and a delay margin of:

$$0.5081/1.7844 = 0.2847 \text{ s}$$

The delay margin is the 0.19 s of Figure 3, and it answers Figure 1: 0.1 s is below it, so that run settles; 0.25 s is above it, so that run grows. With 0.1 s already in the loop, 0.09 s more is left. RO-060's tuned PD gains, $K_p = 0.2$ and $K_d = 0.283$, keep far more phase and delay margin at every delay (notebook):

Loop, delay	Phase margin	Gain margin	Delay margin
ours, no delay	22.6°	no limit	0.19 s
ours, 0.1 s	10.7°	13.4	0.09 s
ours, 0.25 s	unstable	unstable	unstable
tuned, no delay	76.4°	no limit	0.458 s
tuned, 0.1 s	59.7°	5.38	0.358 s
tuned, 0.25 s	34.6°	2.03	0.208 s

The tuned loop crosses over higher, at 2.91 rad/s, but its phase margin is more than three times ours, so it takes more delay. Its gain margin under delay is smaller, because its larger derivative gain keeps the loop's gain higher at the high frequencies where the delay bends the phase to −180°. With its integral gain of 0.05 added, the tuned loop has a phase margin of 75.8° at 2.86 rad/s and a delay margin of 0.463 s.

A delay costs phase in proportion to the frequency. The same delay therefore uses up more of the margin in a loop with a higher crossover frequency. The three margins each measure one direction away from −1; a change of gain and phase together can reach −1 sooner, which the closest distance of the plot to −1 measures, through the sensitivity function.

6.4 The margins on the Bode plot

The same margins can be read off the loop's Bode plot (FBS §10.3), the usual way to read them when designing. The critical point −1 has a gain of 1, which is 0 dB, and a phase of −180°. So in Figure 10: find where the gain curve crosses 0 dB and read how far the phase is above −180° (phase margin); find where the phase curve crosses −180° and read how far the gain is below 0 dB (gain margin). Watch the solid phase curve: the 0.1 s delay bends it down past −180° at 11.66 rad/s, while the dashed delay-free curve never reaches −180°.

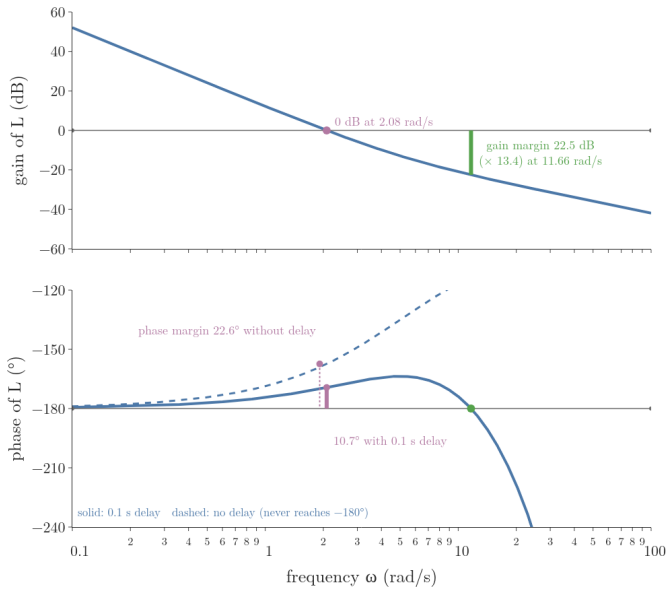


Figure 10: The Bode plot of the heading loop's L: gain (top) and phase (bottom). Solid: with 0.1 s of delay; dashed: without (the gain is the same for both). The gain crosses 0 dB at 2.08 rad/s. Phase there: −157.4° without delay (phase margin 22.6°) and −169.3° with 0.1 s (phase margin 10.7°). With 0.1 s the phase reaches −180° at 11.66 rad/s, where the gain is −22.5 dB: gain margin 22.5 dB, a factor of 13.4. Without delay the phase never reaches −180°

This Bode reading needs the conditions of Section 5: no poles of L in the right half-plane, and the curves crossing 0 dB and −180° only once. When the curves cross several times, the Nyquist plot gives the margins (FBS §10.3).

7. Where margins and delay show up in real stacks

> **Key point:** Real flight and driving stacks treat delay as a direct loss of margin: PX4 trades filtering against delay so that the rate loop's gains can be raised, and Autoware predicts ahead by its 0.17 s actuator delay instead of losing that phase.

PX4, the drone autopilot. PX4's multicopter tuning guide calls the time from a disturbance to the motors' reaction the control latency, and says that "lowering latency allows you to increase the rate **P** gains" (PX4 filter tuning). Its table gives the delay of the gyro's low-pass filter: about 8 ms at a 30 Hz cutoff, 3.8 ms at 60 Hz and 1.9 ms at 120 Hz. The delay margin says why: a delay T_d costs phase ω times T_d at the crossover, and raising the gains moves the crossover up, so the same delay costs more. On our heading loop, crossing over at 2.08 rad/s, an 8 ms filter would cost only 0.95° of the 22.6° (notebook); a loop crossing over ten times higher would lose ten times as much.

Autoware, the self-driving stack. Autoware's PID longitudinal controller drives a car's speed with a PID controller (G-2965), PD control plus a term that adds up the past error.

Its README says that gas and brake actuators “typically” have “a delay of about a hundred millisecond”, and its parameter file sets `delay_compensation_time: 0.17` s (Autoware PID controller). Rather than accept the phase lost to this delay, the controller predicts the car’s speed 0.17 s ahead and feeds the prediction back. For our heading loop, a 0.17 s delay would use 90 percent of its 0.19 s delay margin; for RO-060’s tuned loop it would use 37 percent of 0.458 s and leave a phase margin of 48.0° (notebook).

8. Summary

Idea	Formula or number	Why it matters
Loop transfer function	$L(s) = C(s)G(s)e^{-sT_d}$; ours $(0.8\text{s} + 4)/\text{s}^2$ times the delay	everything about stability is read off L
Closed-loop poles	where $L(s) = -1$; $-0.4 \pm 1.96\text{j}$ without delay	a pole on the right means growing swings
Argument principle	$N = Z - P_f$	turns of the image count zeros minus poles inside a walk
Nyquist criterion	$Z = N + P$	counts unstable closed-loop poles from L alone, delay included
Gain margin	$1/\lvert L(j\omega_{pc})\rvert$; none without delay, 13.4 with 0.1 s	how much the gains can grow
Phase margin	$180° + \angle L(j\omega_{gc})$; 22.6°, 10.7° with 0.1 s	how much phase lag the loop can take
Delay margin	φ_m/ω_{gc}; 0.19 s, 0.09 s more with 0.1 s	how late the heading may arrive

- The heading loop can take 0.19 s of delay before it oscillates, so Figure 1's 0.1 s run settles and the 0.25 s run grows, because a delay T_d subtracts ω times T_d of phase and at 2.08 rad/s, where the loop's gain is 1, 0.19 s uses up all of its 22.6° phase margin; without delay its gains can grow without limit, with 0.1 s of delay 13.4 times.
- The **loop transfer function** (G-3035) L, the product of everything a signal meets once round the loop, decides the closed loop's poles, because they are the points where $L = -1$, the **critical point** (G-3036).
- A sine of 2.08 rad/s that comes back once round the loop exactly as it went in, gain 1 and phase $-180°$, keeps itself going, so $L(j\omega) = -1$ is the edge of stability.
- The **argument principle** (G-3037) counts zeros minus poles inside a closed walk by the **encirclements** (G-3038) of the image, because each zero inside turns the image once clockwise and each pole once counter-clockwise.
- The **Nyquist contour** (G-3039) walks round the whole right half-plane, stepping round the double pole at 0, and the **Nyquist plot** (G-3040) is L drawn along it, so the **Nyquist stability criterion** (G-3041), $Z = N + P$, counts the closed loop's unstable poles: 0 for our 0.1 s loop, 2 for the 0.25 s loop.
- A Bode-plot reading misleads when L has poles in the right half-plane: the balancing robot shows a gain of 2 at $-180°$ yet is stable, because its one unstable pole of L is balanced by one counter-clockwise turn round -1.
- The **gain margin** (G-3044), read at the **phase crossover frequency** (G-3043), is unlimited for the delay-free loop and 13.4 with 0.1 s, because the delay alone bends the phase down to $-180°$.
- The **phase margin** (G-3045), read at the **gain crossover frequency** (G-3042), is 22.6°, below the usual 30° to 60°, so the loop has little room for any extra lag, such as

a filter or a slower computer; RO-060's tuned gains have 76.4°.
- The **delay margin** (G-3046) is the phase margin divided by the gain crossover frequency, 0.19 s for our loop and 0.458 s for RO-060's tuned loop, so a larger phase margin tolerates more delay, while a higher crossover uses it up faster; PX4 cuts filter delay to raise its gains, and Autoware predicts across its 0.17 s actuator delay.

We can now say how much gain, phase lag or delay the heading loop can take, and check stability even for a robot that falls without control. Still missing: each margin measures one direction away from -1, so a combined change can break a loop with good margins, and nothing yet says how the loop treats disturbances and sensor noise; the sensitivity functions measure both.

9. Sources

Built from

- Control System Lectures, "Nyquist Stability Criterion, Part 1", YouTube, https://www.youtube.com/watch?v=sof3meN96MA (Nyquist lecture)
- MATLAB, "Why the Bode Plot Is Awesome!", MATLAB Tech Talk, 2026, YouTube, https://www.youtube.com/watch?v=BGAjnt-C4tI (MATLAB Bode talk)
- Åström, K. J. and Murray, R. M. (2021). *Feedback Systems: An Introduction for Scientists and Engineers*, 2nd ed., Princeton University Press. Ch. 10 "Frequency Domain Analysis": §10.1 The Loop Transfer Function (Example 10.1, the delay $\tau c = (\pi + \theta_0)/\omega_0$), §10.2 The Nyquist Criterion (Theorems 10.1 to 10.3, Example 10.5, the inverted pendulum), §10.3 Stability Margins (eqs. 10.5 and 10.6, typical values, the delay margin); §12.1 Sensitivity Functions. Free: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-loopanal_24Jul2020.pdf (FBS)

Other references

- The MathWorks, `allmargin` documentation (gain, phase and delay margins; its example $L = 25/(s^3 + 10s^2 + 10s + 10)$: phase margin 29.11° at 1.7844 rad/s, delay margin 0.2847 s), https://www.mathworks.com/help/control/ref/dynamicsystem.allmargin.html
- PX4 Autopilot, "MC Filter Tuning & Control Latency", file `docs/en/config_mc/filter_tuning.md`, read for the latency table and the advice on P gains (PX4 filter tuning), https://github.com/PX4/PX4-Autopilot/blob/main/docs/en/config_mc/filter_tuning.md
- Autoware Foundation, PID longitudinal controller, in the folder `control/autoware_pid_longitudinal_controller/`, files `README.md` (time delay compensation) and `config/autoware_pid_longitudinal_controller.param.yaml` (`delay_compensation_time: 0.17`), read from the files (Autoware PID controller), repository https://github.com/autowarefoundation/autoware_universe

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Loop transfer function $L(s)$ (G-3035)	The product of every transfer function a signal meets once round a feedback loop, such as controller times plant times delay, $L = CGe^{-sT_d}$; the closed loop's poles are where $L(s) = -1$, so stability and the margins are read off L alone.
Critical point (G-3036)	The point -1 of the complex plane: wherever the loop transfer function equals -1 the closed loop has a pole, so the Nyquist criterion counts turns round it and the margins measure the distance to it.
Gain crossover frequency ω_{gc} (G-3042)	The frequency at which the loop transfer function's gain is 1 (0 dB), such as 2.08 rad/s for the robot's heading loop; the phase margin and the delay margin are read there.
Encirclement (winding number) (G-3038)	One full turn of a closed curve round a point; the winding number is the net count, clockwise minus counter-clockwise, which the Nyquist criterion reads off the Nyquist plot round -1.
Argument principle (G-3037)	The rule that when a point walks clockwise once round a closed path, the image of a transfer function turns clockwise round 0 once for every zero inside minus once for every pole inside, $N = Z - P$; it lets us count poles in a region without solving for them.
Nyquist contour (G-3039)	The closed path, walked clockwise, up the whole imaginary axis and back round a very large half circle, with very small half circles round any poles on the axis, so it encloses the right half-plane where unstable poles would lie.
Nyquist plot (G-3040)	The loop transfer function drawn as the Nyquist contour is walked: mostly the frequency response, one point per frequency with the gain as length and the phase as angle, plus its mirror image; its turns round -1 decide closed-loop stability.

Term	Meaning
Nyquist stability criterion (G-3041)	The test $Z = N + P$: the closed loop has Z poles in the right half-plane, where N is the number of clockwise turns of the Nyquist plot round -1 and P the number of the loop transfer function's poles there; stable when $Z = 0$, delays and open-loop-unstable plants included.
Phase crossover frequency ω_{pc} (G-3043)	The frequency at which the loop transfer function's phase is $-180°$, where the Nyquist plot crosses the negative real axis; the gain margin is read there.
Gain margin (G-3044)	How many times the loop gain can grow before the closed loop becomes unstable, $1/\lvert L(j\omega_{pc})\rvert$ (often in dB); it says how far the Nyquist plot's crossing of the negative real axis is from -1.
Phase margin (G-3045)	How much extra phase lag the loop can take at its gain crossover frequency before the closed loop becomes unstable, 180ř + $\angle L(j\omega_{gc})$; typical designs keep 30° to 60°.
Delay margin (G-3046)	The smallest extra time delay that makes the closed loop unstable, the phase margin in radians divided by the gain crossover frequency (0.19 s for the robot's heading loop); it says how late sensor readings or commands may arrive.
PD control (G-2962)	Feedback with a proportional part and a derivative part, such as K_p e + K_d de/dt; it acts like a spring and a damper, so it can bring a frictionless joint or heading to its target without swinging.
Step response (G-2981)	The output of a system after a step input, such as the robot's heading after its command jumps to 0.5 rad; its rise time, overshoot, settling time and steady-state error describe how well a loop follows a command.

Term	Meaning
Pole (G-3000)	A value of s where a transfer function's denominator is zero, so the fraction becomes infinite there, such as −2 (twice) for the PD heading loop; each pole adds a term e^(pt) to every response, so its real part says whether the term decays or grows and its imaginary part how fast it oscillates.
Left half-plane (G-3003)	The part of the plane of complex numbers where the real part is negative, left of the imaginary axis; poles there add terms that fade, so a system whose poles all lie there is stable, while the right half-plane holds growing terms.
Frequency response (G-3020)	How a stable linear system responds to a sine at each rate ω once its start-up swing has died away: the gain and the phase, equal to the length and angle of its transfer function at $s = j\omega$; it tells how much of a wave of each rate gets through and how late.
Gain (amplitude ratio) (G-3021)	How far the output sine swings divided by how far the input sine swings at one rate, such as 1.288 for the heading loop at 1 rad/s; it says how much of a wave of that rate the system passes, often written in decibels.
Phase (phase shift) (G-3022)	How far the output sine is shifted against the input sine at one rate, as a fraction of a period in degrees, negative when the output is late, such as −14.9° for a 0.261 s lag in a 6.28 s period; it says how late a system responds to a wave.
Time delay (dead time) (G-3012)	A wait of T seconds between a signal and its effect, such as a camera reporting the heading late, with transfer function e^(−sT); it has no poles or zeros but limits the gains a loop can use, here making the turn swing for ever near 0.32 s.
Proportional gain (G-2959)	The factor K_p by which a proportional controller multiplies the error, such as 0.2 N m per rad or 20 N per m/s; a larger gain gives a stronger correction, and for an undamped plant swings faster.

Term	Meaning
Derivative gain (G-2961)	The factor K_d by which derivative action multiplies the error's rate of change, such as 0.283 N m per rad/s; too little leaves a swing, too much makes the response sluggish.
Moment of inertia (G-2955)	How hard a body is to spin up or slow down about an axis: the sum of each small piece of mass times its distance from the axis squared, such as 0.1 kg m^2 for the robot about its vertical axis; it plays for turning the part mass plays for driving.
Transfer function (control) (G-2999)	The number a linear system multiplies an exponential input e^(st) by, written as a function of s, such as $G(s) = 1/(8s + 2)$ for the robot's speed and $10/s^2$ for its heading; it turns a differential equation into a fraction, so systems can be combined and read by algebra.
Closed-loop transfer function (G-3007)	The transfer function from a closed loop's command to its output, $PC/(1 + PC)$ for a controller C and a plant P, such as $(4s + 10K_i)/(s^3 + 4s^2 + 4s + 10K_i)$ for the PID heading loop; its poles are the loop's poles, shared by every input of the loop.
Natural frequency (G-2991)	The rate, in rad/s, at which a second-order system would swing with no damping, $\omega_n = \sqrt{K_p/I}$, 2 rad/s for the heading loop; it sets how fast the response is, and raising it compresses the whole response in time.
Damping ratio (G-2992)	A second-order system's damping compared with the damping that just stops it from swinging, $\zeta = K_d/(2\sqrt{K_p I})$, 0.2 for D gain 0.08 and 1 for 0.4; it alone sets the shape of the response and its overshoot.
Bode plot (G-3023)	A pair of plots of a frequency response against the rate on a log scale: the gain in decibels and the phase in degrees; on it the pieces of a system add and the response is sketched with straight lines.

Term	Meaning
PID controller (G-2965)	A controller whose command is the sum of a proportional, an integral and a derivative part of the error, K_p e + K_i (integral of e) + K_d de/dt; the commonest feedback controller, it reacts to the error now, its past and its trend; with K_d = 0 it is a PI controller, with K_i = 0 a PD controller.